CODE REVIEW · AUTOMATED CO-PO ATTAINMENT ENGINE

Required changes: backend integration branch

For Abhijeet · from Rohan · 28 Oct 2026 · branch `abhijeet-backend-integration`, commit `9d3c225` (71 files, +5,465 / −158)

> **Status: not ready to merge yet.** Good progress: the backend compiles (Java 21, Spring Boot 4.1.1), the frontend builds, and the MongoDB address now comes from the `MONGODB_URI` environment variable. Items 1 to 4 must be fixed before merging into `main`. Items 5 to 9 are needed before the frontend can switch from demo data to your API. Items 10 to 13 are clean-up.

HOW IT WAS CHECKED

I compiled the backend with `mvnw compile` and ran the frontend development build on your branch: both pass. I trial-merged the branch with the current `main`, and read the code against `frontend/API_CONTRACT.md` (v2.7) and `AGENTS.md`. I didn't run the backend tests, because they need a live MongoDB.

Must fix before merging

MUST FIX **1. Rebase onto the current `main`**

The branch starts from `d7c6405`, so it is missing six later commits on `main`: question paper validation, the marksheet, the README, and Mayuri's question paper AI. Merging it now conflicts in two files:

- `.gitignore`: both branches added one. Keep a single file that has both sets of entries.
- `frontend/src/app/app.routes.ts`: keep the file from `main` as it is, including `step/marksheet` and `step/paper_validation`, which must come before `step/:stepKey`. Then add only your new route.

MUST FIX **2. Course outcomes must not have a Bloom level**

The project rules (AGENTS.md, contract §9) say COs have **no** Bloom level: Bloom levels belong to questions only, and question papers already set them per sub-question. Please remove `bloomLevel` from:

- `obe/CourseOutcome.java:24` and the checks in `obe/CourseOutcomeService.java:52` and `:86`;
- `frontend/src/app/core/api/course-outcome.service.ts`;
- the Bloom dropdown and column in `real-course-overview.component.ts`.

MUST FIX **3. The dashboard calls the live API even in demo mode**

- `faculty-dashboard.component.ts` calls `getRealDashboard()` in its constructor every time, even when `environment.useMockData` is `true`. Without your server running, each dashboard load logs `REAL DASHBOARD ERROR`. Remove the `console.log` and `console.error` calls too.
- `core/api/faculty.service.ts` hard-codes the faculty ID (`6ae0b30416c39e3cbbca0a1c`) and the academic year (`2026-27`). Both must come from the signed-in user (contract FAC-1 and FAC-3 use `/me/…`) and from the request.
- `faculty.service.ts` and `course-outcome.service.ts` hard-code `http://localhost:8081/api/v1`. Use `environment.apiBaseUrl`: the frontend has a single base URL (contract §2.1).

MUST FIX **4. Delete the old `Abhijeet/` folder and change the database password**

- `backend/` now holds the whole project, so `Abhijeet/` is an older duplicate of it. Delete it in the same pull request.
- `Abhijeet/src/main/resources/application.properties` still contains the full Atlas connection string with the password, and that string stays in the git history even after the folder is deleted. Change the database user's password in MongoDB Atlas, then set the new connection string only in your local `MONGODB_URI`. Never commit it, or any `.env` file.

Needed before the frontend can use the API

CONTRACT **5. Put every endpoint under `/api/v1`**

Seven controllers use `/api/…` without the version: `courses`, `departments`, `institutions`, `program-years`, `programs`, `semesters` and `syllabi`. The faculty and CO controllers already use `/api/v1`. The contract's base path is `{API_BASE}/api/v1` for everything (§2.1).

CONTRACT **6. Match the contract's paths and response shapes**

YOURS	CONTRACT
`GET /faculty/{facultyId}/dashboard` returning counts	FAC-3 `GET /me/dashboard` returning `FacultyDashboard` (§7.3): offerings with lifecycle progress, the attention list and the next action
`GET /faculty/{facultyId}/offerings`	FAC-1 `GET /me/offerings` returning `FacultyOffering[]`
`GET /offerings/{offeringId}` returning `RealCourseOffering`	FAC-2, same path, returning `FacultyOffering`
`/programs`, `/syllabi`, `/semesters` …	ACAD-1 to ACAD-9 (§6), e.g. `/semesters/{semesterId}/courses`

When the shapes match, the existing screens work against your API just by setting `useMockData: false`. There is then no need for a separate "real" screen or a second `ApiResponse` type in the frontend.

CONTRACT **7. Return errors in the contract's envelope**

There is no global `@RestControllerAdvice`. The 42 `ResponseStatusException` and 6 `IllegalArgumentException` throws currently return Spring's default error body. The frontend expects `ApiError`: `success: false`, `errorCode`, `message`, optional `fieldErrors`, and `timestamp` (§2.2, error codes in §18).

- Add one global handler, and move the handler that is now inside `CourseOutcomeController` into it.
- Add the optional `errorCode` field to `ApiResponse`.
- Send bad input back as 400 `VALIDATION_FAILED` with `fieldErrors`, and lifecycle locks as 409 `STAGE_LOCKED`.

CONTRACT **8. Sign-in and access checks**

No endpoint checks who is calling. Every endpoint except login and password reset needs `Authorization: Bearer <accessToken>`. Return 401 for a missing or expired token and 403 for a user who isn't allowed (§2.8, §5, roles in §4.3). This doesn't have to land in the same pull request, but the `/me/…` endpoints depend on it.

CONTRACT **9. Keep course outcomes inside the CAM, not a separate screen**

The faculty enters COs in step 1 of the CAM (contract §9), inside the course's ordered lifecycle. The new `/faculty/real-courses/:offeringId` screen and the stand-alone `/course-outcomes` CRUD sit outside that lifecycle. Please build the CAM endpoints (§9.2) instead, so the existing CAM editor saves to MongoDB. After that, remove `real-course-overview.component.ts` and the "Real Assigned Courses" section of the dashboard.

Clean-up

CLEAN-UP **10. Don't reformat files you aren't changing**

`app.routes.ts` and the dashboard component and template were reformatted line by line, which hides the real changes and causes merge conflicts. Keep the existing formatting and change only the lines you need.

CLEAN-UP **11. `MongoConnectionTest` runs on every start**

It lives in `src/main`: it pings the database and prints to the console each time the app starts. Remove it, or move the check into a test (or into a health endpoint).

CLEAN-UP **12. Tests that don't need Atlas**

The only test loads the app context, and it needs a live database. Add service and controller tests against an embedded MongoDB or Testcontainers, starting with the checks listed in contract §10.7.9 and the Bloom range check you wrote.

CLEAN-UP **13. Smaller commits**

One commit holds the whole backend, frontend wiring and a new screen. Please split it into one commit or pull request for the backend modules and one for the frontend wiring.

Suggested order

- Rebase onto `main` (1), remove the Bloom level (2), guard the live calls behind `useMockData` (3), delete `Abhijeet/` and change the password (4), then open a pull request.
- Next pull request: `/api/v1` paths and the error handler (5, 7), then FAC-1 to FAC-3 and ACAD-1 to ACAD-9 in the contract's shapes (6).
- Then sign-in (8) and the CAM endpoints (9).

Contract references are to `frontend/API_CONTRACT.md` v2.7. Questions about a contract shape: ask Rohan before changing it, so the frontend and contract stay in step.